General sexual reproduction in fungi includes three steps 1. Fusion of two protoplasts (plasmogamy) 2. Fusion of nuclei (karyogamy) and 3. Production of haploid spores through meiosis. Methods of reproduction in fungi is given in Figure 1.19.

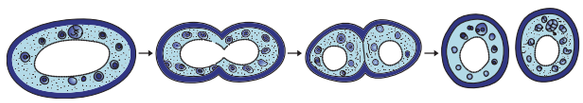
1.5.3 Methods of Reproduction in Fungi

Asexual Reproduction

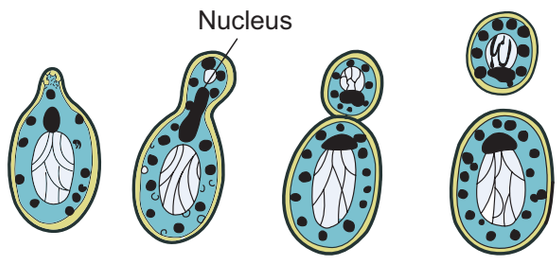
1. Zoospores: They are flagellate structures produced in zoosporangia (Example: Chytrids).
2. Conidia: The spores produced on condiophores (Example: *Aspergillus, Penicillium*).
3. Oidia/Thallospores/Arthrospores: The hypha divided and developed in to spores called oidia (Example: *Erysiphe*).
4. Fission: The vegetative cell divides into 2 daughter cells. (Example: *Schizosaccharomyces*-yeast).
5. Budding: A small outgrowth is developed on parent cell, which gets detached and becomes independent. (Example: *Saccharomyces*-yeast)
6. Chlamydospore: Thick walled resting spores are called chlamydospores (Example: *Fusarium*).

Sexual Reproduction

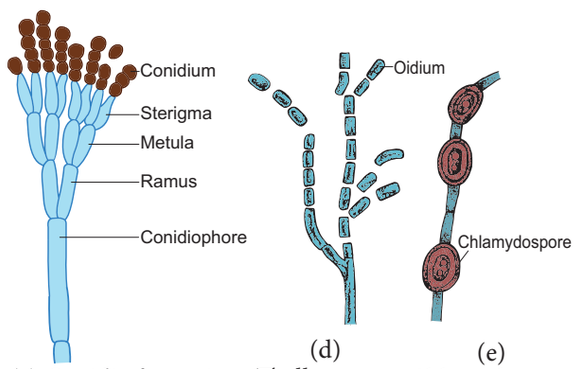
1. Planogametic copulation: Fusion of motile gamete is called planogametic copulation. a. Isogamy – Fusion of morphologically and physiologicall similar gametes. (Example: *Synchytrium)*. b. Anisogamy – Fusion of morphologically or physiologically dissimilar gametes (Example: *Allomyces*). c. Oogamy – Fusion of both morphologically and physiologically dissimilar gametes. (Example: *Monoblepharis*).
2. Gametangial contact: During sexual reproduction a contact is established between antheridium and Oogonium (Example: *Albugo*).
3. Gametangial copulation: Fusion of gametangia to form zygospore (Example: *Mucor, Rhizopus*).
4. Spermatization: In this method a uninucleate pycniospore/microconidium is transferred to receptive hyphal cell (Example: *Puccinia, Neurospora*)
5. Somatogamy: Fusion of two somatic cells of the hyphae (Example: *Agaricus*)



(a) Fission - Yeast



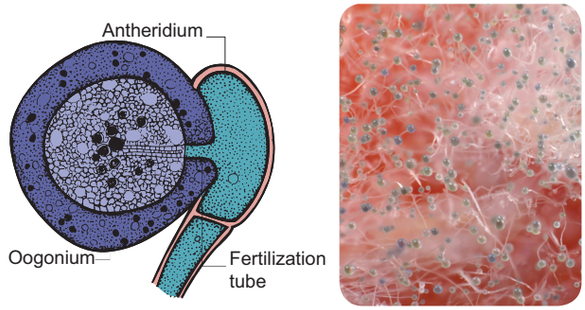
(b) Budding - Yeast



(c) Conidia formation - *Penicillium*

(d) Thallospore - Erysiphe

(e) Chlamydospore - Fusarium



(f) Gametangial contact - *Albugo*

(g) Sporangia - *Mucor*

Figure 1.19: Reproduction in Fungi

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